Tadeusz (Tadek) A.P. Kosmal

Blacksburg, VA — tadekk@vt.edu — github.com/tadekkosmal

Education

Virginia Tech — Blacksburg, USA

Ph.D. Mechanical Engineering — GPA: 4.0/4.0 — 2022 – 2026

Design, Research, and Education of Additive Manufacturing Systems Lab

U.S. Navy SMART Scholar — NSWCCD C618

B.S. Mechanical Engineering — GPA: 3.90/4.0 — 2018 – 2022

Major in Robotics and Mechatronics

Technical Skills

Robotics and Perception: ROS2, RobotStudio, OpenCV, Machine Learning
Programming: Python, C/C++, MATLAB, Git, Linux
Advanced Manufacturing & Design: CAD/CAM, nTopology, KiCAD, Prototyping

Academic Research and Projects

Autonomous, Perception-Based Advanced Manufacturing — *Dissertation*

- Invented a modular in-situ perception array capable of measuring the layer-wise geometric signature of a part. Fused stereo depth with LWIR images to capture thermal history.
- Created an original employment of task-specific geometric priors in stochastic surface estimation to deliver closed, decision manifolds for fabrication action.
- Demonstrated the first, complete perception-based subtractive/hybrid manufacture of components from in-situ, as-built geometric data.
- Realized a multi-axis path planning controller to selectively segment and amend a damaged region from in-situ perception.

Robotic Aerial Vehicle Fabricator — *Principal Investigator*

- Proposed, won, and led a NASA-funded research project with 11 team members as the PI. Pitched and raised an additional $40k material budget from 8 external stakeholders.
- Built a robotic work cell capable of autonomously constructing and deploying customized Unmanned Aerial Systems in under 2 hours, linked here YouTube.
- Spearheaded robotic path planning framework adaptable to changing drone designs, linking CAD design tools and robotic simulation environments.
- Presented and published the work to key stakeholders including: Solid Free Form Symposium 2023, RAPID 2023, NAMRC 2023, ONR/USMC, and more.

Publications, Preprints, & Patents

T. Kosmal et al., "Hybrid additive robotic workcell for autonomous fabrication of mechatronic systems - A case study of drone fabrication," Additive Manufacturing Letters, vol. 3, p. 100100, Dec. 2022, doi: 10.1016/j.addlet.2022.100100.

T. Kosmal et al., **Autonomous Fabrication of Mechatronic Systems**. US Patent App. 18/298,116, 2023

T. Kosmal et al., **Self-Contained, Modular Sensor Aray for In-Situ Monitoring and Data Fusion in Manufacturing**. US Patent App. 19/237,535, 2025

Work Experience

NREIP Intern — NSWC C618 — Summer 2021
Virginia Tech/NSWC Waterborne Drone Intern — Summer 2020